Hantro VC9000E JPEG Encoder

Test Bench User Manual

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Preface

This chapter introduces the *Hantro VC9000E JPEG Encoder Test Bench User Manual*.

About This Document

This document introduces the test bench released for Hantro VC9000E JPEG video encoder IP and provides simulation guidelines.

Additional Reading

- Hantro VC9000E Video Encoder Hardware Features
- Hantro VC9000E JPEG Encoder Hardware Features
- Hantro VC9000E Video Encoder Memory Buffer and Format Organization
- Hantro VC9000E JPEG Encoder API User Manual
- Hantro VC9000E Video Encoder Wrapper Layer API User Manual
- ISO/IEC International Standard 10918-1:1994. Information technology. Digital compression and coding of continuous-tone still images.
- ITU-T Rec. H.265: High-efficiency video coding
- ITU-T Rec. H.264: Advanced video coding for generic audiovisual services
- ITU-R REC. BT.601: Studio encoding parameters of digital television for standard 4:3 and wide-screen 16:9 aspect ratios
- ITU-R REC. BT.709: Parameter values for the HDTV standards for production and international programme exchange
- ITU-R REC. BT.2020: Parameter values for ultra-high-definition television systems for production and international programme exchange

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1 Overview

This chapter gives an overview of the IP core.

1.1 IP Introduction

The VC9000E video encoder is a single-core solution that supports encoding streams with a picture size up to 32768x32768. It features small silicon footprint, low power consumption, and negligible load on the host CPU.

1.2 Supported Codec Standards

The APIs discussed in this version of the document are compatible with the following video codec standards:

- JPEG ITU-T Rec. T.81 (09/92), baseline
- JFIF JPEG File Interchange Format, see `link`

2 Options

This chapter describes various options regarding different features. You can run options in the following way:

./jpeg_testenc [options] -i inputfile

The default value of an option is marked by square brackets in the option description.

2.1 Help Information

Short	Long Option	Description
-H	--help	Display help information.

2.2 Pre-processing Frames

2.2.1 Input Frame Resolutions and Cropping

Short	Long Option	Description
-i[s]	--input	Reads input from file. [input.yuv]
-a[n]	--firstPic	First picture of input file. [0]
-b[n]	--lastPic	Last picture of input file. [0]
-w[n]	--lumWidthSrc	Source image width. [176]
-h[n]	--lumHeightSrc	Source image height. [144]
-x[n]	--width	Output image width. [lumWidthSrc]
-y[n]	--height	Output image height. [lumHeightSrc]
-X[n]	--horOffsetSrc	Horizontal offset of output image. [0]
-Y[n]	--verOffsetSrc	Vertical offset of output image. [0]

Short	Long Option	Description
-W[n]	--write	Whether to write output. [1] 0 - NO 1 - YES

2.2.2 Input Picture Format and Controls

Short	Long Option	Description
-g[n]	--frameType	Input color format. [0] 0 - YUV420P8b_Raster_A16N (YUV 4:2:0 planar(IYUV/I420)) 1 - YUV420SP8b_Raster_A16N (YUV 4:2:0 semi-planar(NV12)) 2 - YVU420SP8b_Raster_A16N (YUV 4:2:0 semi-planar(NV21)) 3 - YUV422YUYV8b_Raster_A16N (YUYV422 interleaved (YUYV/YUY2) 4 - YUV422UYVY8b_Raster_A16N (UYVY422 interleaved (UYVY/Y422) 5 - RGB565_Raster_A16N (RGB565) 6 - BGR565_Raster_A16N (BGR565) 7 - XRGB1555_Raster_A16N (RGB555) 8 - XBGR1555_Raster_A16N (BGR555) 9 - XRGB4444_Raster_A16N (RGB444) 10 - XBGR4444_Raster_A16N (BGR444) 11 - XRGB8888_Raster_A16N (RGB888) 12 - XBRG8888_Raster_A16N (BGR888) 13 - XRGB2101010_Raster_A16N (RGB101010) 14 - XBGR2101010_Raster_A16N (BGR101010) 15 - YUV420P10bWL_Raster_A16N (I010) 16 - YUV420SP10bWH_Raster_A16N (P010) 17 - YUV420P10b_Raster_A16N 18 - YUV420Y0L210b_Raster_A16N 19 - YUV420 customer private tile for HEVC 20 - YUV420 customer private tile for H.264 21 - YUV420SP8b_YuvSp4x4_A16N (YUV 4:2:0 semi-planar tile) 22 - YVU420SP8b_YuvSp4x4_A16N (YUV 4:2:0 semi-planar tile)

Short	Long Option	Description
-g[n]	--frameType	23 - YUV420SP10bWH_YuvSp4x4_A32N (P010 tile) 24 - YUV420SP10bDWL_Raster_A16N (YUV 4:2:0 semi-planar 10) 25 - YUV422SP8b_Raster_A16N (YUV 4:2:2 semi-planar) 32 - YUV420YUV8b_Yuv128x2_A16N (YUV420 UV 8bit tile128x2) 35 - YUV420SP8b_YuvSp8x8_A16N (YUV420 semi-planar UV 8bit tile8x8) 36 - YUV420SP10bWH_YuvSp8x8_A16N (YUV420 semiplanar UV 10bit tile8x8) 37 - YUV420PYVU8b_Raster_A16N (YVU420 UV 8bit planar) 38 - YVU420SP8b_YuvSp64x2_A16N (YUV420 semi-planar UV 8bit tile64x2) 39 - YVU420SP10b_YuvSp128x2_A16N (YUV420 semi-planar UV 10bit tile128x2) 40 - RGB888_Raster_A16N (RGB888) 41 - BGR888_Raster_A16N (BGR888) 42 - RBG888_Raster_A16N (RBG888) 43 - GBR888_Raster_A16N (GBR888) 44 - BRG888_Raster_A16N (BRG888) 45 - GRB888_Raster_A16N (GRB888) 46 - YVU422SP8b_Raster_A16N (YUV 4:2:2 semi-planar) 50 - RGBX8888_Raster_A16N 51 - BGRX8888_Raster_A16N 52 - RGBX1010102_Raster_A16N 53 - BGRX1010102_Raster_A16N 54 - YVU420SP10bWH_Raster_A16N (P010 CrCb) 55 - Y8b_Raster_A16N (YCbCr 4:0:0) Monochrome 8-bit 56 - Y10bWL_Raster_A16N (YCbCr 4:0:0) Monochrome 10-bit occupies 2B [9:0] 57 - Y10bWH_Raster_A16N (YCbCr 4:0:0) Monochrome 10-bit occupies 2B [16:5] 58 - YUV420SP10b_Raster_A16N 62 - YUV422YVYU8b_Raster_A16N (YVYU422) 63 - YUV422VYUY8b_Raster_A16N (VYUY422) 64 - YVU420SP8b_YuvSp8x8_A64N 65 - Y10b_Raster_A16N For more details, see *Frame Buffer Format Specifications*.

Short	Long Option	Description
-v[n]	--colorConversion	RGB-to-YUV color conversion type. [0] 0 - conversion to YUV values without any range change according to Rec. ITU-R BT.601. 1 - conversion to YUV values without any range change according to Rec. ITU-R BT.709. 2 - conversion using custom coefficients. 3 - conversion according to Rec. ITU-R BT.2020. 4 - conversion from full range RGB to limited range YUV according to Rec. ITU-R BT.601. 5 - (just for test) conversion from RGB in range of (0∼219) to limited range YUV according to Rec. ITU-R BT.601. 6 - conversion from full range RGB to limited range YUV according to Rec. ITU-R BT.709.
-G[n]	--rotation	Input image rotation mode. [0] 0 - Disabled 1 - 90 degrees right 2 - 90 degrees left 3 - 180 degrees
-M[n]	--mirror	Whether to enable horizontal mirroring for input image. [0] 0 - Disable 1 - Enable
-Q[n]	--inputAlignmentExp	Alignment value of input frame buffer. [4] 0 - Disable alignment 4 to 12 - Base address of input frame buffer and each line are aligned to 2^inputAlignmentExp
-d[n]	--enableConstChroma	Whether to set chroma to a constant pixel value. [0] Value range: 0 and 1 0 - Disable 1 - Enable
-e[n]	--constCb	Constant pixel value for the U component. [128] Value range: 0 to 255
-f[n]	--constCr	Constant pixel value for the V component. [128] Value range: 0 to 255
	--scanType	The scan type of input image. [0] 0 - raster 1 - supertileX

2.2.3 OSD Overlay Control

A maximum of 12 OSD regions are supported. Two OSD regions cannot share the same MCU.

Short	Long Option (N = 01 to 12)	Description
	--overlayEnables	Overlay region status, with 12 bits representing 12 regions respectively. [0] 1: Region 1 enabled 2: Region 2 enabled 3: Region 1 and 2 enabled and so on.
	--olInputN	Input file for overlay region. [olInputi.yuv]
	--olFormatN	Overlay input format. [0] Value range: 0 to 2 0: ARGB8888 1: NV12 2: Bitmap
	--olAlphaN	Global alpha value for NV12 and bitmap overlay format. [0] Value range: 0 to 255
	--olWidthN	Overlay region width. It can only be set when a region is enabled. [0] It must be under eight-pixel aligned for bitmap format.
	--olHeightN	Overlay region height. It can only be set when a region is enabled. [0]
	--olXoffsetN	Horizontal offset of overlay region top-left pixel. [0] It must be two-pixel aligned. [0]
	--olYoffsetN	Vertical offset of overlay region top left pixel. [0] It must be two-pixel aligned. [0]
	--olYStrideN	Luma stride in bytes. The default value depends on format. Value range: [olWidthi * 4] when the format is ARGB8888. [olWidthi] when the format is NV12. [olWidthi / 8] when the format is bitmap.
	--olUVStrideN	Chroma stride in bytes. The default value depends on the luma stride.
	--olCropXoffsetN	Top left horizontal offset for OSD cropping. [0] For non-bitmap formats, it must be under two-pixel aligned. For bitmap format, it must be under eight-pixel aligned.

Short	Long Option (N = 01 to 12)	Description
	--olCropYoffsetN	Top left vertical offset for OSD cropping. [0] For non-bitmap formats, it must be under two-pixel aligned. For bitmap format, it must be under eight-pixel aligned.
	--olCropWidthN	OSD cropping width. [olWidthi] For bitmap format, it must be under eight-pixel aligned.
	--olCropHeightN	OSD cropping height. [olHeighti]
	--olBitmapYN	Y value of the OSD bitmap format. [0]
	--olBitmapUN	U value of the OSD bitmap format. [0]
	--olBitmapVN	V value of the OSD bitmap format. [0]
	--olSuperTileN	Whether the OSD input data is organized in supertile mode. [0] 0 - non-supertile mode. 1 - X-major supertile mode. 2 - Y-major supertile mode.
	--olScaleWidthN	The width of each OSD input. [0] If a region is enabled, this option must be specified for the region. The option value must be 8 aligned for the bitmap format.
	--olScaleHeightN	The height of each OSD input. [0] If a region is enabled, this option must be specified for the region.

2.2.4 Mosaic Area Control

A maximum 12 mosaic regions are supported. Any region should be aligned to CTB.

Mosaic and OSD region cannot be enabled at the same time. Mosaic region has a higher priority.

For example:

When mosaic region 01 is enabled, OSD region 1 cannot be enabled.

When mosaic region 02 to 12 is enabled, OSD region 1 can be enabled.

Short	Long Option	Description
	--mosaicEnables	Enabling status of mosaic regions, with one bit for one region. (Not support Monochrome encoding if enable) [0] 1: Region 1 enabled 2: Region 2 enabled 3: Region 1 and 2 enabled and so on.
	--mosSizeIndex	different Mosaic size.[0] 0:8x8 1:16x16 2:32x32 3:64x64 4:128x128
	--mosAreaN (N = 01, 02, ..., 12)	Mosaic region parameters, including coordinates of the left, top, right, and bottom pixels. All coordinates must be CTB/MCU aligned.

2.2.5 OSDMap Controls

OSDMap has the lowest priority Comparing with overlay and mosaic. OSDMap and overlay supertile should not be turned on at the same time.

Maximum 12 color attribute (Y[i],U[i],V[i],Alpha[i]) are supported, where i=01,02,..,12.

Each 4bits in OSDMap defines a color attribut applied on one 8x8 block. So 1 byte OSDMap data defines a region of 16x8. osdMapStride should not less than (codingWidth + 15) $>>$ 4.

Following options is example option for color "01". The other color should use corresponding index.

Short	Long Option	Description
	--osdMapEnable	The OSD map enable. [0] 0:disable 1:enable
	--osdMapInput	input file for OSD Map. [NULL]
	--osdMapStride	OSD Map stride in byte. [width]
	--osdMapBlockSize	OSD Map different block size in pixel. [0] 8:8x8 4:4x4
	--osdMapAlphaN	0..255 Specify the alpha value for color[i]. [0]
	--osdMapYN	OSD Map format color[1] Y value. [0]
	--osdMapUN	OSD Map format color[1] U value. [0]
	--osdMapVN	OSD Map format color[1] V value. [0]

2.2.6 DEC400 Tile Status

Short	Long Option	Description
	--dec400TableInput	DEC400 compressed table input from file. [dec400CompTableinput.bin]
	--osdDec400TableInputN	The file from which the video encoder reads the DEC400 compression table for OSD input. [NULL]
	--dec400TileSize	0 - default 1 - 128byte. 2 - 512byte. 3 - 256byte.
	--dec400TileMode	0 - default.[0] 1 - TILE_8x8_x. 2 - TILE_8x8_y. 3 - TILE_16x4. 4 - TILE_8x4. 5 - TILE_4x8. 6 - RASTER_16x4. 7 - TILE_64x4. 8 - TILE_32x4. 9 - RASTER256x1. 10 - RASTER128x1. 11 - RASTER64x1. 12 - TILE_16x8. 13 - RASTER32x4. 14 - RASTER32x1. 15 - RASTER16x1. 16 - TILE_8x4_s. 17 - TILE_16x4_s. 18 - TILE_32x4_s. 19 - TILE_32x8.
	--dec400DataAlignment	0 - default 1 - 32byte. 2 - 64byte. 3 - 1byte. 4 - 16byte.
	--dec400RGBAX	0 - ARGB.[0] 1 - XRGB.

Short	Long Option	Description
	--dec400RGBFormat	0 - default 1 - RGB8. 2 - RGB10. 3 - RGB4. 4 - RGB1555. 5 - RGB565.
	--dec400TSHeaderEnable	Whether to process dec400 tile status header buffer.(128byte) 0 - disable 1 - enable

2.2.7 UFBC Parameters

Short	Long Option	Description
	--ufbcMode	The core mode of UFBC. 0 - disable 1 - afbc (only support 32x8 pixels) 2 - afbc (support 32x8 and 16x16 pixels) 3 - dec400 4 - pvric
	--ufbcYuvTrans	Whether to enable interal YUV transformation. 1 - enable 0 - disable
	--ufbcBlockType	The size of the superblock. Version 1.x 0 - 32x8 pixels 1 - 16x16 pixels Version 2.x 0 - 8x8 pixels 1 - 16x4 pixels 2 - 32x2 pixels
	--ufbcBlockSplit	Whether to enable the superblock split mode for version 1.x. 0 - disable 1 - enable
	--ufbcConstantVal	The constant color for version 2.x.

2.3 Coding Tools and Syntax Control

2.3.1 Coding Modes and Tools

Short	Long Option	Description
-R[n]	--restartInterval	Restart interval in MCU rows. [0]
	--quality	Set quality factor instead of qLevel. [-1] -1 - use quantization table defined by qLevel or fixedQP. 1..100 - use quantization table generated accordingly. Value 100 stands for the best quality.
-q[n]	--qLevel	Quantization scale. [1] Value range: 0 to 10 The value 10 indicates the testbench-defined quantization table is used.
	--qTableFile	External quantization file to be read when quantization level is 10. The file has 16 lines separated by space to specify two quantization tables, with the first 8 lines indicating luma table, and the rest 8 lines indicating chroma table. Each line has 8 values ranging from 1 to 255.
-p[n]	--codingType	Encoding type. [0] 0 - Whole frame encoding 1 - Partial frame encoding
-m[n]	--codingMode	Encoding mode. [0] 0 - YUV420 1 - YUV422 (only for input YUV422SP and YVU422SP) 2 - Monochrome
-t[n]	--markerType	Quantization/Huffman table markers. [0] 0 - Single marker 1 - Multiple markers

2.3.2 APP0 information

Short	Long Option	Description
-u[n]	--units	Unit of X- and Y-density. [0] 0 - Pixel aspect ratio 1 - Dots per inch 2 - Dots per centimetre
-k[n]	--xdensity	X-density to APP0 header. [1]

Short	Long Option	Description
-l[n]	--ydensity	Y-density to APP0 header. [1]
-I[s]	--inputThumb	Reads thumbnail input from file. [thumbnail.jpg]
-T[n]	--thumbnail	Thumbnail to stream. [0] 0 - NO 1 - JPEG 2 - RGB8 3 - RGB24
-K[n]	--widthThumb	Thumbnail output image width. [32]
-L[n]	--heightThumb	Thumbnail output image height. [32]
-c[n]	--comLength	Comment header data length. [0]
-C[s]	--comFile	Comment header data file. [com.txt]

2.3.3 Lossless Encoding

Short	Long Option	Description
	--lossless	Lossless encoding mode. [0] Value range: 0 to 7 0 - Disable 1 to 7 - Enable, with selected prediction modes 1 to 7
	--ptrans	Point transform value for lossless encoding. [0] Value range: 0 to 7

2.3.4 ROI Map

Short	Long Option	Description
	--roimapFile	Input file for ROI map region. [NULL] NULL - Disable ROI Map. "roimap.roi" - text file name to describe ROI regions
	--nonRoiFilter	Input file for non-ROI map region filter. [NULL] NULL - NO exteranl filter, use pre-defined filter level "filter.txt" - specify external filter
	--nonRoiLevel	Non-ROI filter level [5] Value range: 0 to 9, from strong to weak.

2.3.5 Motion JPEG and Rate Control

Short	Long Option	Description
-J[n]	--mjpeg	Whether to enable motion JPEG [0] 0 - Disable 1 - Enable
-B[n]	--bitPerSecond	Target bits per second. [0] 0 - Disable RC None-zero value - RC enabled with the specified target bits per second.
-n[n]	--frameRateNum	Output frame rate numerator. [30] Value range: 1 to 1048575
-r[n]	--frameRateDenom	Output frame rate denominator. [1] Value range: 1 to 1048575
-V[n]	--rcMode	JPEG/MJPEG RC mode. [1] Value range: 0 to 2 0 - Single frame RC mode. 1 - Video RC with CBR. 2 - Video RC with VBR.
-U[s]	--picQpDeltaRange =[Min:Max]	QP delta range in picture-level rate control. Min: Minimum Qp_Delta in picture RC. [−2] Value range: −10 to −1 Max: Maximum Qp_Delta in picture RC. [3] Value range: 1 to 10 The value ranges only apply to two neighboring frames.
-E[n]	--qpMin	Minimum frame QP. [0] Value range: 0 to 51
-F[n]	--qpMax	Maximum frame QP. [51] Value range: 0 to 51
-O[n]	--fixedQP	Fixed QP for every frame. [−1] Value range: −1 to 51 −1 - Disable fixed QP mode. 0-51 - Fixed QP value.

2.3.6 SRAM Control

Short	Long Option	Description
	--sramPowerdownDisable	Whether to disenableSRAM power down [0] 0 - Enable sram power down 1 - Disable sram power down

Short	Long Option	Description
	--sramPowerdownMode	SRAM hw model [0] 0 - Mode 00 1 - Mode 01 2 - Mode 10 3 - Mode 11

2.4 Stream Output Control

Short	Long Option	Description
-o[s]	--output	Writes output to file. [stream.jpg]
	--streamBufChain	Stream buffer output modes. [0] 0 - Single output stream buffer. 1 - Two output stream buffers chained together. **Note**: Minimum size of the first stream buffer is 1KB plus the thumbnail data size.
	--hashtype	The method to calculate the hash value of the output stream. [0] 0 - hash value calculation disabled 1 - standard CRC32 2 - 32-bit checksum

2.4.1 Stream Multi-Segment for Output Low Latency

Short	Long Option	Description
	--streamMultiSegmentMode	Stream multi-segment mode. [0] Value range: 0 to 3 0 - Disable 1 (Reserved) - Enable (hardware handshaking, loop-back enabled) 2 (Reserved) - Enable (software handshaking, loop-back enabled) 3 - Enable (IRQ only, no loop-back)
	--streamMultiSegmentSize	Segment size in byte. [1024] Value range: 512 byte to 16KB It should be a multiple of 16 bytes.

Short	Long Option	Description
	--streamMultiSegmentAmount	Total amount of segments. [4] Value range: 2 to 1024 0 - Full image size = Width x Height x 2

2.5 Input Low Latency Mode

Short	Long Option	Description
-S[n]	--inputLineBufferMode	Input line buffer mode. [0] Value range: 0 to 4 0 = Disable 1 = Enable (software handshaking, loopback enabled) 2 = Enable (hardware handshaking, loopback enabled) (effective only when the upstream IP is used, only VCE IP cannot be tested) 3 = Enable (software handshaking, loopback disabled) 4 = Enable (hardware handshaking, loopback disabled) (effective only when the upstream IP is used, only VCE IP cannot be tested)

Short	Long Option	Description
-N[n]	--inputLineBufferDepth	Number of MCU rows to control loop-back or handshaking [1] Value range: 0 to 511 - When loop-back is enabled, each of the two continuous ping-pong input buffers contains MCU rows specified by this option. - When hardware handshaking is enabled, handshaking signal is processed per CTB/MCU rows specified by this option. - When software handshaking is enabled, IRQ is sent and Read Count Register is updated each time a number of MCU rows specified by this option have been read. **Note**: This option can be set to 0 only when inputLineBufferMode is set to 3. In this case, IRQ is not sent and Read Count Register is not updated.
-s[n]	--inputLineBufferAmountPerLoopback	Line buffer amount in the case of buffer read/write address loopback. [0] Value range: 0 to 1023
	--segmentUnitHeight	Segment unit height when low latency is in SBI mode. [16] Value range: 8 and 16
	--inputSliceInfoEn	Control DDR low-latency mode. [0] 0 - Disable 1 - Enable **Note**: HW will poll slice info structure saved in DDR, which in a 64 byte ddr space

2.6 Hardware Control

2.6.1 AXI Interface

Short	Long Option	Description
	--AXIAlignment	AXI alignment setting (in hexadecimal format). [0] bit[31:28] AXI_burst_align_wr_common bit[27:24] AXI_burst_align_wr_stream bit[23:20] AXI_burst_align_wr_chroma_ref bit[19:16] AXI_burst_align_wr_luma_ref bit[15:12] AXI_burst_align_rd_common bit[11:8] AXI_burst_align_rd_prp bit[7:4] AXI_burst_align_rd_ch_ref_prefetch bit[3:0] AXI_burst_align_rd_lu_ref_prefetch
	--burstMaxLength	Maximum AXI burst length. [16]

2.6.2 IRQ Type

Short	Long Option	Description
	--irqTypeMask	IRQ type mask setting (in binary format). [111110000] swreg1 bit24 irq_type_sw_reset_mask, default 1 swreg1 bit23 irq_type_fuse_error_mask, default 1 swreg1 bit22 irq_type_buffer_full_mask, default 1 swreg1 bit21 irq_type_bus_error_mask, default 1 swreg1 bit20 irq_type_timeout_mask, default 1 swreg1 bit19 irq_type_strm_segment_mask, default 0 swreg1 bit18 irq_type_line_buffer_mask, default 0 swreg1 bit17 irq_type_slice_rdy_mask, default 0 swreg1 bit16 irq_type_frame_rdy_mask, default 0

2.6.3 Peripherals Control (C-Model Only)

Short	Long Option	Description
	--useVcmd	(Only valid fo CModel) Whether to enable VCMD. 0 - Disable 1 - Enable

2.6.4 Low latency gating

Short	Long Option	Description
	--lowlatGatingDisable	0..1 Lowlatency handshake auto gating disable[1] 0: enable lowlatency handshake auto gating. 1: disalbe lowlatency handshake auto gating. If enable, check emc and recon idle.

2.7 Debugging

2.7.1 Dump Registers

Short	Long Option	Description
	--dumpRegister	Whether to dump register. [0] Value range: 0 - Dump no register. 1 - Dump print register. When it is set to 1, software dumps register values to stdout or a log file according to the `--logOutDir` setting.

2.7.2 Log and Print Message Control

Short	Long Option	Description
	--logOutDir	Log message output directory. [0] 0 - All logs output to stdout. 1 - One log file for all logs. 2 - One log file of each instance thread.
	--logOutLevel	Log message output level. [3] 0 -"quiet", no output 1 - "fatal", serious errors 2 - "error", error occurs 3 - "warn", warning 4 - "info", general information 5 - "debug", debug information 6 - "all", all log information

Short	Long Option	Description
	--logTraceMap	Trace information of log message output for prompting and debugging. [63]=b`0111111 Bit 0 - Dump API call. Bit 1 - Dump registers. Bit 2 - Dump EWL. Bit 3 - Dump memory usage. Bit 4 - Dump Rate Control Status Bit 5 (This bit is reserved for future use) - Output full command line Bit 6 - Output performance information

2.7.3 Testing

These options are not supported for end users.

Short	Long Option	Description
-P[n]	--trigger	Logic analyzer trigger for the nth picture. [−1] −1 - Disable the trigger.
-D[n]	--XformCustomerPrivateFormat	Customer private format to be converted from YUV420. [−1] −1 - Disable conversion 0 - Customer private tile format for HEVC 1 - Customer private tile format for H.264 2 - Customer private YUV422SP_888 3 - Common data 8-bit tile 4x4 4 - Common data 10-bit tile 4x4 5 - Customer private tile format for JPEG
	--osdDec400TableInput	OSD DEC400 compressed table input from file. [osdDec400CompTableinput.bin]

2.7.4 vcmd priority and core bit mask

Short	Long Option	Description
	--priority	The priority of current instance in vcmd mode. [0] 0 - normal priority 1 - high priority

Short	Long Option	Description
	--core_mask	The core bit mask of current instance. [0] 0 - not specify, anycore. 1 - bit[0]=1: specify core 0. 2 - bit[1]=1: specify core 1.

3 Input Color Formats Mappings to FFmpeg and Microsoft-defined Formats

Table 3.1 lists typical input formats and their mappings to FFmpeg formats and formats defined by Microsoft. For details about the color formats, see *Hantro VC9000E Series Memory Buffer and Format Organization*.

Table 3.1 Mappings of Input Color Formats to FFmpeg and Microsoft-defined Formats

--inputFormat	Description	FFmpeg Format	Format Defined by Microsoft
0	YUV420P8b_Raster_A16N (IYUV/I420)	yuv420p	IMC4
1	YUV420SP8b_Raster_A16N (NV12)	nv12	NV12
2	YVU420SP8b_Raster_A16N (NV21)	nv21	-
3	YUV422YUYV8b_Raster_A16N (YUYV/YUY2)	yuyv422	YUY2
4	YUV422UYVY8b_Raster_A16N (UYVY/Y422)	uyvy422	UYVY
5	RGB565_Raster_A16N	rgb565le	-
6	BGR565_Raster_A16N	bgr565le	-
7	XRGB1555_Raster_A16N	rgb555le	-
8	XBGR1555_Raster_A16N	bgr555le	-
9	XRGB4444_Raster_A16N	rgb444le	-
10	XBGR4444_Raster_A16N	bgr444le	-
11	XRGB8888_Raster_A16N	bgr0/bgra	-
12	XBRG8888_Raster_A16N	rgb0/rgba	-
40	RGB8888_Raster_A16N	bgr24	-
41	BGR8888_Raster_A16N	rgb24	-

--inputFormat	Description	FFmpeg Format	Format Defined by Microsoft
15	YUV420P10bWL_Raster_A16N (I010)	yuv420p10le	-
16	YUV420SP10bWH_Raster_A16N (P010)	p010le	P010

4 Pre-Processing Features

4.1 RGB-to-YUV Conversion

The optional color space conversion (CSC) block reads the data of each picture in 30-, 24-, 16-, 15-, or 12-bit RGB. It then converts the picture data into planar YUV420 format by sub-sampling chrominance samples vertically and rearranging the samples.

The input RGB pixel is converted into YUV by using the following equations:

$$Y' = A \times r + B \times g + C \times b$$
$$Y = clip(Y' + LumaOffset)$$
$$U = clip(E \times b - G \times Y' + Const)$$
$$V = clip(F \times r - H \times Y' + Const)$$

where:

- A, B, C, E, F, G, and H are color conversion coefficients in Q0.16 fixed-point format.
- LumaOffset is the color conversion offset, which is supported only if the value of `EWLHwConfig_t.cscExtendSupport` is `1`.
- Const is the constant offset for chroma, which is 128 for 8-bit encoding and 512 for 10-bit encoding.
- clip() is the function that clips the result into the range of corresponding bit depth.

When input format is RGB, the CSC function will be enable to convert RGB input as YUV for encoding. the conversion transform will be speicified using option "--colorConversion".

4.2 Picture Rotation

The picture rotation block rotates each picture 180 or 90 degrees clockwise or counterclockwise. The following figure illustrates the resultant encoded picture dimensions after rotation.

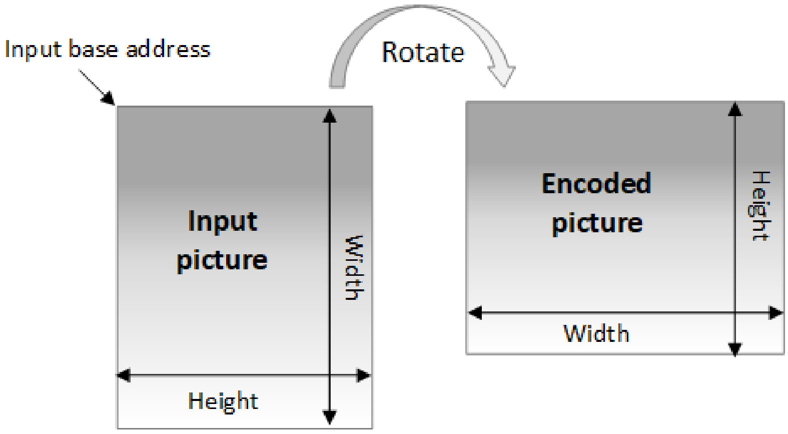

Figure 4.1 Picture Dimensions After Rotation

The picture rotation feature is supported only if the value of `EWLHwConfig_t.NonRotationSupport` is `0`. When rotation is supported, you can control it by using the option `-rotation` in testbench. function.

4.3 Picture Cropping

The picture cropping block allows reading and encoding a selected portion of a picture.

The crop area to be encoded is specified by setting the size of the input picture, the size of the crop area, and the top-left corner coordinates of the crop area relative to the top-left corner of the input picture, as shown in the following figure.

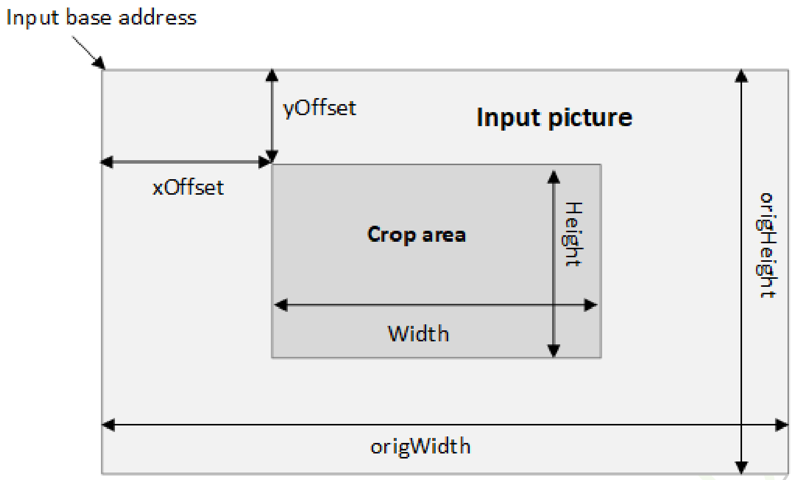

Figure 4.2 Input Picture and Crop Area

The picture cropping block requires the following conditions on the width, height, and their offsets of the crop area:

$$origWidth - xOffset - width \geq 0$$

$$origHeight - yOffset - height \geq 0$$

The input picture horizontal stride in pixels must be a multiple of 2. For example, when the input width is 639 pixels, the horizontal stride must be 640 pixels.

4.4 On-the-Fly OSD Mixer

The video encoder supports on-screen display (OSD) for blending input pictures with OSD regions. The generated blended picture is fed into the encoder as final input.

OSD regions are rectangular regions with OSD input in a certain format. The video encoder regards OSD regions as a foreground to blend OSD regions with the input picture based on the following alpha blending algorithm. OSD regions should not overlap or share CTB due to hardware restrictions.

$$result\_pixel = \begin{cases} overlay\_pixel & Alpha == 255 \\ (overlay\_pixel * Alpha + background\_pixel * (256 - Alpha)) >> 8 & Alpha \neq 255 \end{cases}$$

This feature is available only if the value of `EWLHwConfig_t.OSDSupport` is `1`.

4.5 Osdmap

OSD Map is a buffer specified by the user, where every 4 bits correspond to a pixel's color and alpha value (up to 12 colors and alpha values). Then the user can mix the specified area with alapha and the original image according to the specified color in the Map Buffer, and the mixed result is used as the input for encoding. This function can be used to scramble any area. For example, in car applications, color blocks can be applied to the facial part. In the past, the mosaic function could only support up to 12 regions of mosaic in VCE, and using OSD Map did not have this limitation.

The Data in OSD Map buffer are defined as following:

- Each 8x8 block corresponds to a 4-bit control information. The value of these 4-bits determines the color on this 8x8 stack.
- The MSB of one byte defines the block before the block defined in the LSB.

The specific 4-bit control information encoding is as follows:

- 4'd0: Do not stack any information
- 4'd1: Overlay the first color and alpha configuration
- 4'd2: Overlay the second color and alpha configuration
- 4'd8: Overlay the eighth color and alpha configuration
- 4'd12: Overlay the twelfth color and alpha configuration
- 4'd13∼4'd15: Invalid

Note:

- Comparing with overlay and mosaic, OSDMap has the lowest priority.
- OSDMap and overlay supertile should not be both turned on at the same time.
- The sum of available colors in the OSD area and OSDMap cannot be greater than 12, because OSDMAP and OSD share registers to represent the corresponding colors in the Map.
- It is not recommended to open both OSDMAP and Mosaic functions simultaneously, because the goals of OSDMAP and Mosaic functions are similar.
- You can get information on whether this feature is supported from EWLHwConfig_t.OSDMapSupport. When this value is 1, it indicates support.

5 Features and Command Line Examples

This chapter provides the introduction on hantro jpeg encoder and provide the command line example to verify realated feature.

5.1 Set Quantization Parameters

Hantro JPEG encoder's quantization table (DQT) are generated by software and value of the table are send to hardware by a set of registers. Two tables are need to specified, one for luma and another for chroma. User can specify their own quatization table beside the pre-defined ones.

option "-q index" or "--qpLevel index" are used to specify which pre-defined table are used. we can select 10 tables. the quality of table index 0 has worst quality and table 10 has best quality.

```
./jpeg_testenc -w352 -h288 -i ~/vec/foreman_cif.yuv -q8 -b0 -o stream.jpg
```

In addition, table 10 can be modified by external text file as customered quant table. user need to specify the customized quantization table of option "--qTableFile=quantfile". Following is an example of external quant matrix file quant.txt

```
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
```

```
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
```

In above quantization table file, 16 lines are needed. each line a 8 value in range [1:255]. The starting eight lines specify the 64 quantization value for luma, and the following eight lines specify the 64 value for chroma. and we can encode using this table as following example,

```
./jpeg_testenc -w352 -h288 -i ~/vec/foreman_cif.yuv -q10 --qTableFile=quant.txt -b0 -o stream.j
```

5.2 ROI Map for JPEG

When encode a jpeg file, some region are not interested by user and we can encode them with less bits. Hantro JPEG encoder can support such feature by specify a buffer to indicate which blocks are not intested by user. such buffer is called ROI Map buffer in our design.

Note:

- You can get information on whether this feature is supported from EWLHwConfig_t.JpegRoiMapSupport. When this value is 1, it indicates jpeg ROI encoding is supported.

Option "--roimapFile=<regions.roi>" will specify an external file which describe the region user NOT interested in. Here is content of file "roimap.roi" which used in our example.

```
#pic=0;
roi=(16,16,64,64)
roi=(96,80,80,80)
```

each roi line will specify a region by four value (left,top,width,height). It defines two regions:

- rectangle which (left, top) position is (16,16) with (widht, height) of (64,64)
- rectangle which (left, top) position is (96,80) with (widht, height) of (80,80)

Becuase the unit of ROIMAP for JPEG is 16x16 block, the left/right and top/bottom edge will be aligned to 16-pixel position by extending rectangle defined. The test bench will parse the ROI file and generate the data used in ROI Map buffer.

Option "--nonRoiFiliter=<filter.txt>" will specify the filter which will drop the quality of non-roi region. It has two 8x8 matrix for luma and chroma seperatedly. Here is the content of file "filter.txt" which is used in out example.

```
#luma
255 255 255 255 0 0 0 0
255 255 255 255 0 0 0 0
255 255 255 255 0 0 0 0
255 255 255 255 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
#chroma
255 255 255 0 0 0 0 0
255 255 255 0 0 0 0 0
255 255 255 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
```

In this example, the top-left 4x4 part of the DCT coefficient will be kept and the other coefficients will be droped.

In our testbench, some pre-defined filters are integrated. Option "--nonRoiLevel level" is used for it. "level" will determine the abvoe "filter.txt" matrix, we can set 10 levels (from 0 to 9). if don't use this parameter, the filter coefficient will use it in "filter.txt".

```
# use pre-defined filter 5
./jpeg_testenc --input=foreman_cif.yuv --output=stream.jpg --width=352 --height=288 \
 --firstPic=0 --lastPic=0 --roimapFile=roimap.roi  --nonRoiLevel=5
# use external filter defined in "filter.txt"
./jpeg_testenc --input=foreman_cif.yuv --output=stream.jpg --width=352 --height=288 \
 --firstPic=0 --lastPic=0 --roimapFile=roimap.roi  --nonRoiFiliter=filter.txt
```

If successful, the image quality of the non-Roi area will be reduced, and the ROI area will remain unchangedfilter.

5.3 Motion JPEG Encoding

Motion JPEG is a video compression format in which each video frame is compressed separately as a JPEG image. From this point, motion jpeg is a kind of container to save a sequence of JPEG image. In our test bench, beside generate jpeg file one by one and cat them together defined as mjpg container, we integrate a simple AVI container to store the sequence JPGE image.

To encode a sequence use mjpg raw format with a fixed qunat matrix,

```
./jpeg_testenc -w352 -h288 -i ~/vec/foreman_cif.yuv -q9 -b300 -o stream.mjpg
```

To encode a sequence with AVI container, we need to add "--mjpeg=1" option and name the output_iterator stream with "avi" postfix. In addition, we support bitrate control for AVI container. Here is an examples

```
./jpeg_testenc -w352 -h288 -i ~/vec/foreman_cif.yuv --mjpeg=1 \
         --bitPerSecond=1000000 --rcMode 1 -b300 -o stream.avi
```

AVI is a container format that can contain both audio and video data in a file to allow synchronous audio-with-video playback. The postfix "avi" is necessary for generating AVI streams.

Compared with H.264 encoder, compression ratio is larger when using JPEG encoder. It is recommended to set larger bit rate during JPEG encoding to achieve better quality streams.

5.4 JPEG Lossless Process

Lossless process is defined in JPEG specification T.81 Annex H. It uses some pre-defined prediction mode to predict the pixel value directly without doing DCT transform. So the compression ratio is about 1/2 in general condition. The simplified jpeg lossless encoder diagram is as following (from ITU-T.81),

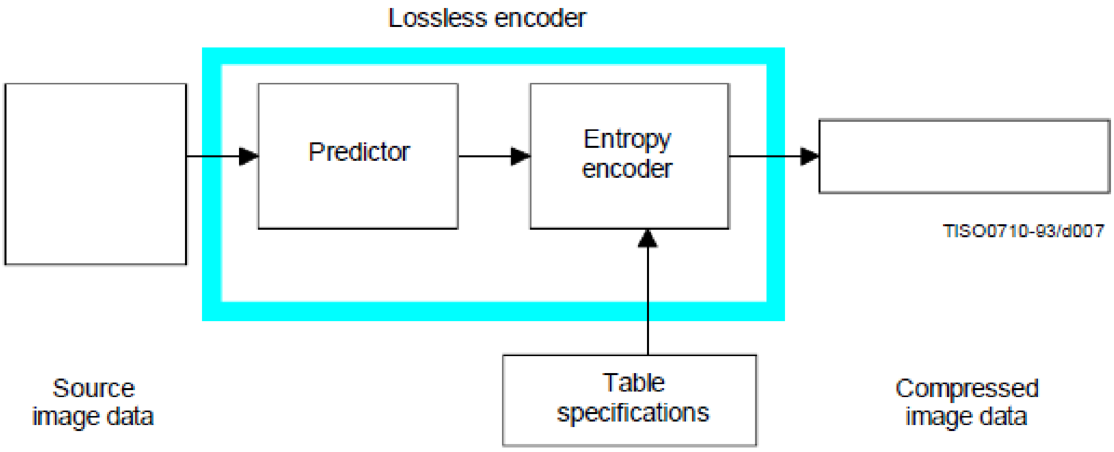

Figure 5.1 JPEG Lossless Encoder Simplified Diagram

Following figure shosw the relationship between the position (a, b, c) of the reconstructed neighboring samples used for predication and the position of x, the smaple being coded. (from ITU-T.81, figure H.1)

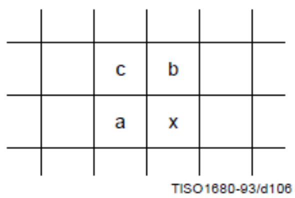

Figure 5.2 Relationship between sample and prediction sampels

Seven predication mode are defined for JPEG lossless process. See following table for prediction algorithm to get Px from the value of Ra, Rb, and Rc. (from ITU-T.81, table H.1)

Selection-value	Prediction
0	No prediction (See Annex J)
1	Px = Ra
2	Px = Rb
3	Px = Rc
4	Px = Ra + Rb – Rc
5	Px = Ra + ((Rb – Rc)/2)[a]
6	Px = Rb + ((Ra – Rc)/2)[a]
7	Px = (Ra + Rb)/2
[a] Shift right arithmetic operation	

Figure 5.3 Prediction for lossless coding

Compared with baseline process, lossless process do sample predicting and do not apply DCT transform. The residual sample after prediction is encoded by entropy encoder (VLC) directly. The same huffman table are used for lossless process.

Point transform is defined to get the sample's value with different precision. It is defined as a shift operation to process the input samples. Point transform parameter defines the shift bit number used. For lossless process, point transform parameter are defined in SOS (start of scan marker).

Note: JPEG lossless process is defined in ITU-81. It is a format not widely used except in some medical image processing system. JPEG LS is another lossless/near-lossless compression standard defined in ITU-T.87, it is different from JPEG Lossless process used here.

In testbench, option "--lossless N" and "--ptrans N" are used for lossless encoding.

- when "--lossless" is not zero, lossless process is enble. the value of lossless is also specified the prediction mode. valid prediction mode is 1 to 7.
- "--ptrans" option defines the point transform parameter. When the value is zero, the input sample vlaue is used directly. or the input sample value is shifted as "ptrans" bits firstly. Then shift result is used to predict.

Following command will encode a lossless jpeg file.

```
./jpg_testenc -i foreman_cif.yuv -w352 -h288 -b0 --lossless=3 \
              --ptrans=0 -o stream.jpg
```

JPEG lossless is not used widely, so most jpeg viewer don't support it. To check decoded result of lossless jpeg, you can use ffmpeg to decode it or use ffplay to display it.

Note:

- You can get information on whether this feature is supported from EWLHwConfig_t.ljpegSupport. When this value is 1, it indicates jpeg lossless encoding is supported.

5.5 Slice Encoding by Set Restart Interval

JPEG doesn't define slice. We use restart interval feature to implement slice-like encoding. So when we mention slice encoding in JPEG, restart interval is used. In addition, in slice mode encoding, after encoding one slice (a restart interval), the encoder will stop. After prepare the input picture of next slice, user can start to encode next slice. In such way, when DDR memory is limited, we can encode the whole picture in several steps. and in each step, we can only encode one slice of the picture.

Restart interval defines a group of MCU which can be processed independently by encoder or decoder. One parameter for restart interval is the number of MCU in this interval. Because the length of the parameter is defined as 16 bit, the max number of MCU in one restart interval is 65535. In large pictures or when MCU is small (e.g. 1x1 for lossless monochrome jpeg), please pay attention that larger restart interval is not valid.

In API, slice encoding is enable when coding type is "JPEGENC_SLICED_FRAME". And parameter JpegEncCfg.restartInterval to specfy how many MCU to encode as a restart internal. For DCT baseline process, the unit of restartInterval is MCU row. For lossless process, the unit of restartInterval is 16 lines.

In testbench, specify "--codingType=1" and "--restartInterval=N" can enable the slice encoding. When specified non-zero value of restartInterval but with "--codingType=0", restart interval is enable, but encoder will not stop after one restart interval is encoded. The last slice maybe not as large as the rows specified in restartInterval when the picture height is not integer times of the restartInterval MCU rows.

Following command will enable slice encoding and encode the picture slice by slice. Each slice is 2 MCU rows. for the picture height is 288 rows, 9 (=288/(16∗2)) slices are encoded.

```
./jpg_testenc -i foreman_cif.yuv -w352 -h288 -b0 --codingType=1 \
              --restartInterval=2 -o stream.jpg
```

Appendix: Glossary

The following table describes the terms used in this document. For additional HEVC acronyms, see http://www.hevcbook.de/acronyms/.

Term	Description
1080p	high-definition resolution of 1920 x 1080 progressive video
720p	high-definition resolution of 1280 x 720 progressive video
ABR	average bit rate
AIoT	artificial intelligence of things
APB	Advanced Peripheral Bus
API	application programming interface
ARIB	Association of Radio Industries and Businesses
ASIC	application-specific integrated circuit
AV1	AOMedia Video 1
AVC	Advanced Video Coding (H.264)
AXI	Advanced eXtensible Interface
bps	bits per second
CABAC	context-adaptive binary arithmetic coding
CAVLC	content-adaptive variable-length coding
CB	coding block
CBR	constant bit rate
CIF	Common Interchange Format (352 x 288 pixels)
CIR	cyclic intra refresh
CPB	coded picture buffer
CPU	central processing unit
CQP	constant quantization parameter
CRF	constant rate factor
CSC	color space conversion
CTB	coding tree block
CTU	coding tree unit

Term	Description
CU	coding unit
CVBR	constrained variable bit rate
DDR	double data rate
DRM	digital rights management
EFBC	external frame buffer compression
EOS	end of sequence
EWL	encoder wrapper layer
FPS	frames per second
GDR	gradual decoder refresh
GOP	group of pictures
HD	high definition
HDTV	high-definition television
HEIF	High Efficiency Image File Format
HDR	high dynamic range
HEVC	High Efficiency Video Coding, a video coding standard (H.265) developed by JCT-VC
HRD	hypothetical reference decoder
IDR	Instantaneous Decoder Refresh
IPCM	intra pulse code modulation
ITU	International Telecommunication Union
ITU-T	ITU Telecommunication Standardization Sector
ITU-R	ITU Radiocommunication Sector
Kbps	kilobits per second
LCU	largest CU
LTR	long-term reference
MAD	mean absolute difference
MB	macroblock
MMU	memory management unit
MPEG-4	Moving Picture Experts Group 4
MV	motion vector
NAL	network abstraction layer
NALU	NAL unit, the smallest decodable unit of a stream
OBU	open bitstream unit
OSD	on-screen display
PAL	Phase Alternate Line (720 x 576 pixels resolution)
POC	picture order count
PPS	picture parameter set
PSNR	peak signal to noise ratio

Term	Description
PU	prediction unit
QCIF	Quarter CIF (176 x 144 pixels)
QVGA	Quarter Video Graphics Array (320 x 240 pixels)
QP	quantization parameter
RC	rate control
RDO	rate-distortion optimization
RDOQ	rate-distortion optimized quantization
REE	rich execution environment
RFC	reference frame compression
RGB	red, green, blue color space representation
ROI	region of interest
RPS	reference picture set
SAO	sample adaptive offset
SBI	segment buffer interface
SEI	supplemental enhancement information
SMPTE	Society of Motion Picture and Television Engineers
SPS	sequence parameter set
SSIM	structural similarity index measure
STR	short-term reference
sub-QCIF	Sub Quarter CIF (128 x 96 pixels resolution)
SVC-T	temporal scalable video coding
SXGA	Super Extended Graphics Array (1280 x 1024 pixels resolution)
TU	transformation unit
TV	television
VBR	variable bit rate
VBV	video buffer verifier
VGA	Video Graphics Array (640 x 480 pixels resolution)
VLC	variable-length code
VPS	video parameter set
VUI	video usability information
YCbCr	color space with one luma and two chroma components, used for digital encoding
YUV	color space with one luma and two chroma component, used for either digital or analog encoding

Document Revision History

This section describes top level differences in the versions of this document.

NOTE: This document is not necessarily updated for each patch or minor revision. The information in this document tends to be stable across a revision (nnn) series.

Document Revision	Date	Compatible Core	Description
1.02	2024-01-02	VC9000E	Updated the document name and template.
1.01	2023-06-12	VC9000E	- Reconstructed the document and refined descriptions. - Added a new input color format, XBGR2101010_Raster_A16N, whose ID is 14, in Section *Pre-processing Frames*. - Updated the stream segment buffer feature.
1.00	2021-03-15	VC9000E	Initial release.